

A Study: Speed Stat Ceiling Breaks

Powercreep makes raising the stat ceiling inevitable. By releasing new units with additional stats

Finding All Ceiling Breaks

Let's find all ceiling breaks, who did it, when, and what version.

```
In [6]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [7]: table = pd.read_csv('../data/lv40_table.csv')
date = table.sort_values(by='release')
breaks = pd.DataFrame(columns=['hero', 'spd', 'release', 'version'])
```

```
In [8]: date
```

Out[8]:

	hero	entry	move	weapon	hp	atk	spd	def	res	total	...	aided	duo	tempest	r
0	Abel: The Panther	Shadow Dragon / (New) Mystery	Cavalry	Blue Lance	39	33	32	25	25	154	...	False	False	False	
197	Chrom: Exalted Prince	Awakening	Infantry	Red Sword	47	37	25	31	17	157	...	False	False	False	
801	Lyn: Lady of the Plains	The Blazing Blade	Infantry	Red Sword	37	28	37	26	29	157	...	False	False	False	
210	Clarine: Refined Noble	The Binding Blade	Cavalry	Colorless Staff	35	25	33	22	29	144	...	False	False	False	
1128	Saizo: Angry Ninja	Fates	Infantry	Colorless Dagger	36	29	34	33	16	148	...	False	False	False	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
682	Laegjarn: Flame Miracle	Heroes	Flying	Green Axe	40	48	49	47	16	200	...	False	False	False	
1181	Sharena: Sea of Hope	Heroes	Infantry	Red Sword	40	46	49	38	31	204	...	False	False	False	
549	Höör: Twilight Breeze	Heroes	Flying	Colorless Bow	40	48	33	24	49	194	...	False	False	False	
87	Baldr: Eternal Shimmer	Heroes	Flying	Green Tome	40	48	49	28	24	189	...	False	True	False	
18	Alfonse: Sunlit Spirit	Heroes	Cavalry	Blue Lance	41	49	16	49	41	196	...	False	False	False	

1389 rows × 32 columns

```

In [9]: fastest = date[date['release'] == date.loc[0]['release']].sort_values(by='spd').iloc[-1]['spd']
breaks.loc[0] = [date[date['release'] == date.loc[0]['release']].sort_values(by='spd').iloc[-1]['release'],
for i in range(date.shape[0]):
    row = date.iloc[i]
    if row['spd'] == fastest:
        if row['release'] == breaks.loc[breaks.shape[0] - 1]['release']:
            breaks.loc[breaks.shape[0]] = row['hero'], fastest, row['release'], row['version']
    elif row['spd'] > fastest:
        fastest = row['spd']
        breaks.loc[breaks.shape[0]] = row['hero'], fastest, row['release'], row['version']
breaks = breaks.drop(0)

```

```

In [10]: print(breaks.to_markdown())

```

	hero	spd	release	version
1	Lon'qu: Solitary Blade	39	2017-02-02	1
2	Mia: Lady of Blades	40	2017-11-15	1.8
3	Mareeta: The Blade's Pawn	41	2019-05-10	3.5
4	Mareeta: Sword of Stars	42	2019-12-06	4
5	Freyja: Lady of Nightmare	44	2020-11-27	4.11
6	Corrin: Starry Seer	45	2022-10-06	6.1
7	Nino: Spirited Sorcerer	46	2023-01-10	7.1
8	Ursula: Royal-Blue Crow	46	2023-01-10	7.1
9	Bernadetta: Late Hibernator	47	2023-03-07	7.3
10	Karla: Spring Reveries	47	2023-03-07	7.3
11	Bernadetta: Sheltered Loner	48	2024-08-16	8.8
12	Felix: Sword Savant	48	2024-08-16	8.8
13	Sharena: Hopeful Princess	49	2025-08-15	9.8

We get this simple table above that lists each ceiling break chronologically, starting with **Lon'qu: Solitary Blade**, who held the highest speed stat upon the release of the game.

Column	Description
hero	Name and epithet of hero
spd	Speed stat of hero
release	Release date of hero
version	Version hero was released in

Analysis

A Skip in Stats?

Something that I immediately noticed was that **43 speed was completely skipped over?? Freyja: Lady of Nightmare** (Mythic) went straight to **44 speed** when she released.

So who was the first 43 speed unit, and when did they appear?

```
In [11]: first43 = date[date['spd'] == 43].iloc[0]
print(f'{first43.hero}\n{first43.release}')
```

```
Katarina: Torchlit Wanderer
2021-01-18
```

The first 43 speed hero, which was **Katarina: Torchlit Wanderer** appeared around two months after the first 44 speed unit.

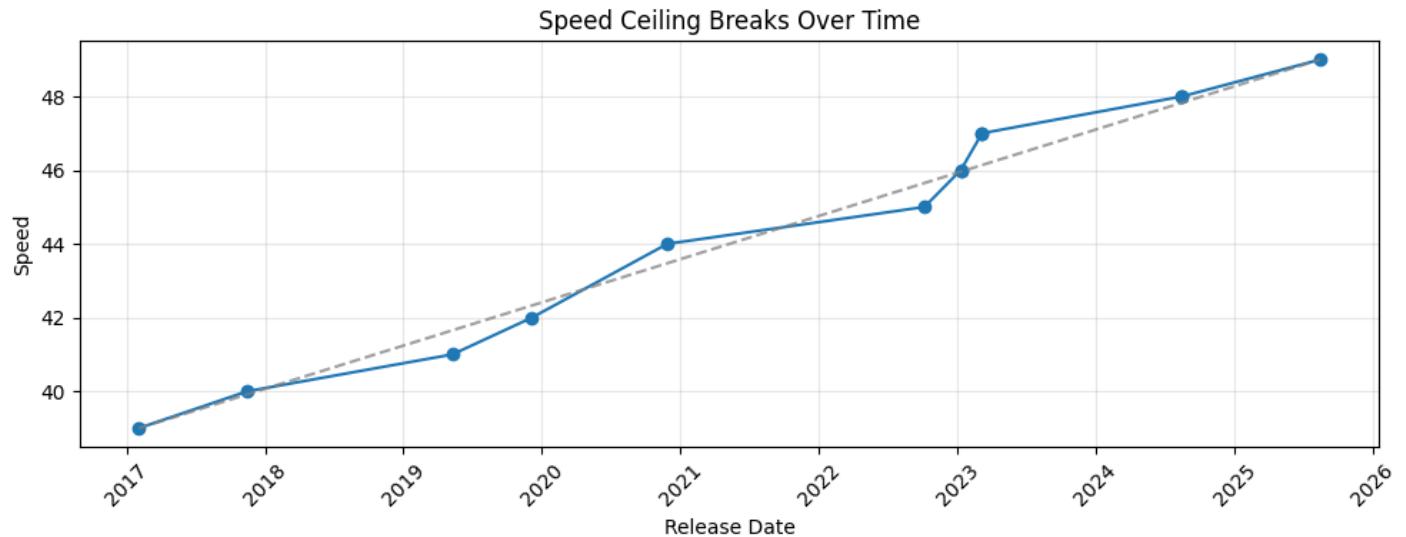
Release Dates

So there's no guarantee that a year or major version will see a speed ceiling break.

Also, there doesn't seem to be a clear pattern to the time between ceiling breaks.

```
In [12]: breaks["release"] = pd.to_datetime(breaks["release"])
```

```
plt.figure(figsize=(10, 4))
plt.plot(breaks["release"], breaks["spd"], marker="o")
plt.plot([breaks["release"].iloc[0], breaks["release"].iloc[-1]],
         [breaks["spd"].iloc[0], breaks["spd"].iloc[-1]],
         linestyle='--', alpha=0.75, color='gray')
plt.title("Speed Ceiling Breaks Over Time")
plt.xlabel("Release Date")
plt.ylabel("Speed")
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



(Note that the slopes of each segment do not signify that the speed stat was slowly increasing over time for units. This is simply to visualize the speed at which the next ceiling was broken.)

What about significance? Is there any significance to the versions or release dates of ceiling-breakers?

Ursula: Royal-Blue Crow and **Nino: Spirited Sorcerer** were units on the yearly desert-themed special banner in January, specifically for the 2023 Khadein banner (and so was **Katarina: Torchlit Wanderer**, who was on the 2022 Plegian banner).

The last two breaks were done by **Sharena: Hopeful Princess** and **Bernadetta: Sheltered Loner**. Both are significant in that they are **Choose Your Legends** units. Specifically, they were the first place winners of the female division for CYL 9 and CYL 8, respectively. Could we say that the next speed break will happen with the release of the CYL 10 units? Maybe. It'd be interesting.

Weapon Type

Let's first get the rest of the data for the units in `breaks`.

```
In [13]: breaks = breaks[['hero']].merge(table, on='hero')
breaks.head()
```

Out[13]:

	hero	entry	move	weapon	hp	atk	spd	def	res	total	...	aided	duo	tempest	regi
0	Lon'qu: Solitary Blade	Awakening	Infantry	Red Sword	45	29	39	22	22	157	...	False	False	False	
1	Mia: Lady of Blades	Path of Radiance	Infantry	Red Sword	38	32	40	28	25	163	...	False	False	False	
2	Mareeta: The Blade's Pawn	Thracia 776	Infantry	Red Sword	42	33	41	29	23	168	...	False	False	False	
3	Mareeta: Sword of Stars	Thracia 776	Infantry	Red Sword	41	35	42	31	23	172	...	False	False	False	
4	Freyja: Lady of Nightmare	Heroes	Cavalry	Green Beast	40	38	44	24	22	168	...	False	False	False	

5 rows × 32 columns

In [14]:

breaks

Out[14]:

	hero	entry	move	weapon	hp	atk	spd	def	res	total	...	aided	duo	tempest	re
0	Lon'qu: Solitary Blade	Awakening	Infantry	Red Sword	45	29	39	22	22	157	...	False	False	False	
1	Mia: Lady of Blades	Path of Radiance	Infantry	Red Sword	38	32	40	28	25	163	...	False	False	False	
2	Mareeta: The Blade's Pawn	Thracia 776	Infantry	Red Sword	42	33	41	29	23	168	...	False	False	False	
3	Mareeta: Sword of Stars	Thracia 776	Infantry	Red Sword	41	35	42	31	23	172	...	False	False	False	
4	Freyja: Lady of Nightmare	Heroes	Cavalry	Green Beast	40	38	44	24	22	168	...	False	False	False	
5	Corrin: Starry Seer	Fates	Infantry	Blue Breath	40	42	45	36	30	193	...	False	False	False	
6	Nino: Spirited Sorcerer	The Blazing Blade	Infantry	Red Tome	38	41	46	20	32	177	...	False	False	True	
7	Ursula: Royal-Blue Crow	The Blazing Blade	Cavalry	Green Tome	40	42	46	18	22	168	...	False	False	False	
8	Bernadetta: Late Hibernator	Three Houses	Cavalry	Blue Bow	39	43	47	20	25	174	...	False	False	True	
9	Karla: Spring Reveries	Heroes	Cavalry	Green Axe	40	45	47	24	22	178	...	False	False	False	
10	Bernadetta: Sheltered Loner	Three Houses	Cavalry	Blue Bow	40	45	48	24	28	185	...	False	False	False	
11	Felix: Sword Savant	Three Houses	Infantry	Red Sword	40	47	48	35	29	199	...	False	False	False	
12	Sharena: Hopeful Princess	Heroes	Cavalry	Green Tome	40	46	49	28	22	185	...	False	False	False	

13 rows × 32 columns

What does the distribution look across color? Across weapon types?

In [15]:

print(breaks['color'].value_counts().to_markdown())

color	count
:	:
Red	6
Green	4
Blue	3

No Colorless representation, unfortunately.

```
In [16]: print(breaks['weapon'].value_counts().to_markdown())
```

weapon	count
:	:
Red Sword	5
Blue Bow	2
Green Tome	2
Blue Breath	1
Green Beast	1
Red Tome	1
Green Axe	1

Pretty on point, I suppose? For god infantry units, Godswords are a favored build. Even after **Saber: Dune Walker** dropped with his **Jehanna Lance**, I saw Godswords more often than Godlances (completely anecdotal).

But, an analysis on the most common infantry archetypes is something for another time. Maybe I'll link it here when I get to it one day.

Considering this, though, I'd say one would naturally suppose that breakers mainly consist of infantry units. Is this true?

Movement Type

Let's do a quick check.

```
In [17]: print(breaks['move'].value_counts().to_markdown())
```

move	count
:	:
Infantry	7
Cavalry	6

An even split! It makes sense that no armored units broke the ceiling, as they aren't exactly a viable speed-build type with most God skills being infantry-locked, specifically for melee weapons.

```
In [18]: print(breaks[['hero', 'move']].to_markdown())
```

	hero	move
---	:-----	:-----
0	Lon'qu: Solitary Blade	Infantry
1	Mia: Lady of Blades	Infantry
2	Mareeta: The Blade's Pawn	Infantry
3	Mareeta: Sword of Stars	Infantry
4	Freyja: Lady of Nightmare	Cavalry
5	Corrin: Starry Seer	Infantry
6	Nino: Spirited Sorcerer	Infantry
7	Ursula: Royal-Blue Crow	Cavalry
8	Bernadetta: Late Hibernator	Cavalry
9	Karla: Spring Reveries	Cavalry
10	Bernadetta: Sheltered Loner	Cavalry
11	Felix: Sword Savant	Infantry
12	Sharena: Hopeful Princess	Cavalry

I'm not sure if this means anything, but breakers used to be infantry units, but now, this trend shifted to cavalry. Perhaps a cav unit, with its extra movement over infantry, seems more appealing than an infantry unit that would need some tempest status or skill, which may require a sacrifice in another skill or stat?

Hero Availability

Is there any significance to how accessible these heroes are?

```
In [19]: def bettervis(cond):
          if cond:
              return '0'
          return ' '

temp = breaks[['hero', 'regular_4_3_2_1', 'specialRate', 'regular_5', 'mythic', 'special']].copy()
for i in temp.columns[1:]:
    temp[i] = temp[i].apply(bettervis)
print(temp.to_markdown())
```


	hero	regular_4_3_2_1	specialRate	regular_5	mythic
special					
----	-----	-----	-----	-----	-----
:	:	:	:	:	:

0	Lon'qu: Solitary Blade	0			
1	Mia: Lady of Blades		0		
2	Mareeta: The Blade's Pawn		0		
3	Mareeta: Sword of Stars		0		
4	Freyja: Lady of Nightmare				0
5	Corrin: Starry Seer		0		
0					
6	Nino: Spirited Sorcerer		0		
0					
7	Ursula: Royal-Blue Crow		0		
0					
8	Bernadetta: Late Hibernator				
0					
9	Karla: Spring Reveries				
0					
10	Bernadetta: Sheltered Loner			0	
11	Felix: Sword Savant			0	
12	Sharena: Hopeful Princess			0	

Breaks started with only normal pool units. After **Freyja: Lady of Nightmare** breaks, the following breakers are all seasonals (incentive for players to pull limited heroes?). At least, until Choose Your Legends units become breakers. Nothing too significant? Their reasoning for certain units being breakers is up for speculation. There is incentive to pull limited units because of their limited availability and likely new skills. Same goes for CYL units that always bring something new to the table in terms of skills and effects.

To Sum Up

These are just a few short question I had in regards to speed ceiling breaks. This is moreso an extension of what I tried doing before, where I tried to predict WIP